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Device for measuring a change in length

Abstract

A device for measuring a change in length has a first fastening element, a second fastening element and at least one length element which is arranged between the two fastening elements. The one length element has a first end a second end and a length along a longitudinal direction. A force acting parallel to the longitudinal direction leads to a change in length of the length element. A lever element has a first end, a second end, and a fulcrum and is arranged transversely to the longitudinal direction. The lever element includes a first lever arm with a first length between the fulcrum and a first lever arm end and a second lever arm with a second length between the fulcrum and a second lever arm end, with the second length being greater than the first length.

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Background/Summary

- (1) This is a National Phase Application filed under 35 U.S.C. 371 as a national stage of PCT/EP2020/085103, filed Dec. 8, 2020, an application claiming the benefit of German Application No. 10 2019 135 732.2 filed Dec. 23, 2019, the content of each of which is hereby incorporated by reference in its entirety.
- (2) The invention relates to a device for measuring a change in length according to the generic term of claim 1.
- (3) In automation technology, 6-axis force torque sensors are frequently used to determine the forces and moments being exerted in all directions. For example, such sensors can be used for automatic joining or assembly of workpieces, for deburring, polishing or grinding, during haptic measurements or other areas of application. These sensors measure the forces (F_x , F_y , F_z) and moments (M_x , M_y , M_z) being exerted on them in and about three coordinates (x , y , z). On the one hand, such sensors should be as rigid as possible so that they themselves do not experience any deformations due to the forces or moments. On the other hand, it should be possible to achieve the highest possible resolution of the signals being measured, which as a rule is not possible with rigid sensors. In the case of known sensors, strain gauges are used to determine forces and moments, with which strain gauges very small material strains can be measured. It is, however, very complex to apply strain gauges. In addition, strain gauges require major amplification of signals, which leads to high costs of such 6-axis force torque sensors.
- (4) The task of the invention is therefore to provide a device for measuring a change in length, in particular for use in 6-axis force torque sensors, which with a high degree of rigidity can achieve a high resolution of the measured signals.
- (5) The task is solved according to the invention by a device for measuring the change in length with the features of patent claim 1.
- (6) Advantageous embodiments and further developments of the invention are indicated in the dependent claims.
- (7) The device for measuring a change in length according to the invention with a first fastening element, a second fastening element and at least one length element which is arranged between the two fastening elements and has a first end, a second end and a length along a longitudinal direction, wherein a force acting parallel to the longitudinal direction leads to a change in length of the length element, is characterized in that a lever element with a first end, a second end and a fulcrum is arranged transversely to the longitudinal direction, wherein the lever element has a first lever arm with a first length between the fulcrum and a first lever arm end and a second lever arm with a

second length between the fulcrum and a second lever arm end, wherein the second length is greater than the first length, and in that the length element is pivotably arranged with its first end at the first lever arm end of the first lever arm and wherein the second lever arm end of the second lever arm is connected to a material measure, the movement of which can be detected by a scanning element.

(8) In other words, the basic idea of the invention is to mechanically amplify the change in length by a lever element and to detect the change of path at the free end of the longer lever arm. A mechanical amplification of this type is easy and inexpensive to achieve. It can, in particular, be used for systems with a high degree of rigidity, wherein notwithstanding the rigidity, a high resolution of the change in length caused by a force or a moment can be achieved by means of the mechanical amplification.

(9) In a preferred embodiment of the invention, the fulcrum is arranged between the first end and the second end of the lever element, and the first lever arm end of the first lever arm forms the first end of the lever element, whereas the second lever arm end of the second lever arm forms the second end of the lever element. In this way, a two-sided lever is formed.

(10) In an alternative preferred embodiment of the invention, the fulcrum is arranged at the first end of the lever element. In this way, a one-sided lever is formed.

(11) A preferred further development of the invention provides that a second element, which has a first end and a second end and which, by means of a first arm and a second arm which are aligned parallel to each other, is arranged parallel to the lever element, wherein the arrangement is pivotable between the second element and the first arm, between the second element and the second arm, between the lever element and the first arm, and between the lever element and the second arm. In this way, a parallelogram is formed, which parallelogram makes it possible for the material measure arranged at the second lever arm end of the second lever arm, which material measure without the second element tilts against the scanning element during a pivot movement of the lever element, to be guided aligned parallel to the longitudinal direction during a pivot movement of the lever element by the second element.

(12) Preferably, the material measure is arranged on the second arm, wherein it preferably extends over the distance between the lever element and the second element. This enables the material measure to be aligned in any position parallel to the longitudinal direction during a pivot movement of the lever element.

(13) Good stability of the lever arrangement can be achieved if, preferably, the first arm is arranged between the fulcrum and the first end of the second element, and the second arm is arranged between the second end of the lever element and the second end of the second element.

(14) A particularly preferred embodiment of the invention provides that the pivotable connections are in the manner of a film hinge. Even in highly rigid materials, such pivotable connections can easily be formed by weakened material regions in such a way that on the one hand the desired high rigidity of the length element is either not or only minimally affected, whereas on the other hand the desired high resolution can be achieved by the mechanical reinforcement by the lever arrangement.

(15) Preferably, the first fastening element is disk-like with a first plane and the second fastening element is disk-like with a second plane, wherein the first plane and the second plane are arranged parallel to each other and preferably the longitudinal direction is arranged perpendicular to the planes. The disk-like design of the fastening elements enables good fastening to the components that are to be moved relative to each other, between which the occurring forces and moments are to be measured.

(16) The arrangement of the planes perpendicular to the longitudinal direction may simplify the determination of the forces and moments being exerted.

(17) A particularly preferred embodiment provides that several, in particular at least six, for example precisely six, length elements are arranged between the first fastening element and the

second fastening element. This enables a determination of the forces and moments being exerted between the two fastening elements in and about three axes and thus the formation as a 6-axis force torque sensor.

(18) Advantageously, the scanning element is implemented as an optical, capacitive, inductive or magnetic scanning sensor. Optical sensors are, in particular, especially robust and enable scanning with high resolution.

(19) Preferably, an evaluation unit is provided which evaluates the signals detected by the at least one scanning element and, in particular, calculates the forces and moments being exerted between the two fastening elements by means of the changes in length.

Description

(1) The invention is elucidated in detail making reference to the following figures. Wherein,
(2) FIG. 1 shows a perspective illustration of a first embodiment example of a device for measuring a change in length according to the invention,

(3) FIG. 2 shows a further perspective illustration of a part of the device according to FIG. 1,

(4) FIG. 3 shows a longitudinal cross-section through the device according to FIG. 2,

(5) FIG. 4 shows a schematic illustration of the lever arrangement of the device according to FIG. 1,

(6) FIG. 5 shows a schematic illustration of a lever arrangement of an alternative embodiment example of a device for measuring a change in length, and

(7) FIG. 6 shows a perspective illustration of a device for measuring a change in length with six length elements according to FIG. 1 to form a 6-axis force torque sensor.

(8) FIGS. 1 to 4 show different views of a first embodiment example of a device **10** according to the invention for the measurement of a change in length ΔL to a length element **15**, which length element has a first end **15a**, a second end **15b**, and a length L along a longitudinal direction R . A force F acting parallel to the longitudinal direction R leads to a change in length ΔL of the length element **15**. A direct measurement of this change in length ΔL to the length element **15** is costly or can often only be detected with low resolution due to the minimal change in length ΔL . The illustrated device **10** therefore comprises a lever element **20** arranged transversely to the longitudinal direction R and having a first end **20a**, a second end **20b** and a fulcrum D , wherein the lever element **20** has a first lever arm **21** with a first length a between the fulcrum D and a first lever arm end **21a** and a second lever arm **22** with a second length b between the fulcrum D and a second lever arm end **22b**. The second length b is formed in a way to be larger than the first length a , wherein the length element **15** is pivotally arranged with its first end **15a** at the first lever arm end **21a** of the first lever arm **21**, whereas the second lever arm end **22b** of the second lever arm **22** is connected to a material measure **30**. The pivotable arrangement between the length element **15** and the first lever arm end **21a** of the first lever arm **21**, may, in particular, be implemented in the manner of a film hinge. The film hinge may be formed, for example, by a weakened material region.

(9) The device **10** further comprises a scanning element **40** that can detect a movement of the material measure **30**.

(10) A force F that acts on the length element **15**, parallel to the longitudinal direction R and bringing about a change in length ΔL to the lever arm end **21a** of the short lever arm **21** causes a change of path Δs to the second lever arm end **22b** of the long lever arm **22** due to the lever element **20**. In so doing, the change in length ΔL is, in particular, amplified at a ratio of $b:a$, the change in length Δs at the second lever arm end **22b** thus amounts to $\Delta s = b \cdot \Delta L / a$.

(11) The change of path Δs is detected by means of the scanning element **40**, which scans the material measure **30** connected to the second lever arm end **22b**. The material measure **30** can, in

particular, be scanned optically, for which purpose, for example, the material measure **30** can be implemented to be reflective.

(12) As illustrated in FIGS. **1** to **4**, in this embodiment, the fulcrum **D** is arranged between the first end **20a** and the second end **20b** of the lever element **20**, such that a two-sided lever is formed. In so doing, the first lever arm end **21a** of the first lever arm **21** forms, in particular, the first end **20a** of the lever element **20** and the second lever arm end **22b** of the second lever arm **22** forms, in particular, the second end **20b** of the lever element **20**.

(13) As illustrated in FIG. **5**, it is possible that the fulcrum **D** is arranged at the first end **20a** of the lever element **20**. This forms a one-sided lever. The two lever arms **21**, **22** extend from the fulcrum **D** in the same direction, but with respectively different lengths *a* or alternatively *b*.

(14) The material measure **30** can be arranged on the front side of the lever element **20**. The scanning element **40** is aligned in such a way that it points in the direction of the end face of the second lever arm end **22b** in order to be able to scan the material measure. In this, the scanning element **40** is especially rigidly fixed, in particular parallel to the longitudinal direction **R** and fixedly relative to the length element **15**. When the lever element **20** moves, the material measure **30** does however pivot around the fulcrum **D** along a circular path, which can lead to distorted measurement signals in the scanning element **40**.

(15) Therefore, irrespective of whether the lever element is implemented as a one-sided or two-sided lever, it is preferable that a second element **50** is provided, which second element has a first end **50a** and a second end **50b**, and which second element is arranged parallel to the lever element by means of a first arm **51**, which has a first end **51a** and a second end **51b**, and by means of a second arm **52**, which is aligned parallel to the first arm **51** and has a first end **52a** and a second end **52b**. The first end **51a** of the first arm **51** is, in particular, connected with the fulcrum **D** of the lever element **20** and the second end **51b** of the first arm **51** is connected with the first end **20a** of the second element **50**, whereas the first end **52a** of the second arm **52** is connected with the second end **20b** of the lever element **20** and the second end **52b** of the second arm **52** is connected with the second end **50b** of the second element **50**. In this, the arrangements between the second element **50** and the first arm **51**, the second element **50** and the second arm **52**, the lever element **20** and the first arm **51**, and between the lever element **20** and the second arm **52** are implemented to be pivotable, in particular, in the manner of a film hinge. The film hinge can be formed, for example, by a weakened material region.

(16) The material measure **30** may be arranged on the second arm **52** and may preferably extend over the distance between the lever element **20** and the second element **50**. This arrangement causes the material measure **30** to continue to be connected to the lever element **20**, but preferably pivotally, such that when the lever element **20** moves due to a force **F** acting on the length element **15**, the material measure **30** continues to be moved in accordance with the transmission ratio, however by means of the guidance of the second element a parallel displacement is brought about such that the material measure **30** remains in its orientation relative to the scanning element **40** as it moves.

(17) The scanning element **40** may comprise or be connected to an evaluation unit **60**, which evaluates the signals detected by the scanning element **40**, and in particular calculates the force **F** acting on the length element **15**, from the changes in lengths **DI** that are determined.

(18) In order to be able to fixedly arrange the scanning element **30** relative to the length element **15**, two cross members **17a**, **17b** are arranged on the length element **15** at the first end **15a** and the second end **15b**, which cross members are guided past the lever arrangement formed by the lever element and optionally the second element and the two arms **51**, **52**, in particular engaging around them on both sides, such that the scanning element can be arranged at the ends of the cross members **17a**, **17b** facing away from the length element **15**.

(19) In order to be able to determine the change in length ΔL upon exertion of a force between two components, the length element **15** is connected at its first end **15a** to a first fastening element **11**

and at its second end **15b** to a second fastening element **12**. If the first fastening element **11** is connected to a first component (not shown) and the second fastening element **12** is connected to a second component (not shown), the forces then being exerted can be determined when the two components move relative to each other. The first fastening element **11** may be formed by the first cross member **17a**, the second fastening element **12** may be formed by the second cross member **17b**.

(20) FIG. 6 shows a perspective view of a device **10'** for measuring a change in length ΔL , which comprises six length elements **15** according to the embodiment example shown in FIGS. 1 to 4. In this, the device **10'** comprises a first fastening element **11'** and a second fastening element **12'**, wherein the cross members **17a** of the six length elements **15** are connected to the first fastening element **11'** and the cross members **17b** of the six length elements **15** are connected to the second fastening element **12'**. The fastening elements **11'**, **12'** are formed in a disc-like manner, for example in the manner of a disc-ring, with a first plane **E1** or alternatively a second plane **E2**, wherein the planes **E1** and **E2** are arranged parallel to one another and, in particular, are arranged perpendicular to the longitudinal direction **R** of the length elements **15**. The length elements **15** are uniformly distributed over the circumference of the fastening elements **11'**, **12'** and, in particular, form a 6-axis force torque sensor. For this purpose, the signals detected by all six scanning elements **40** are fed to the evaluation unit **60**, from which the forces F_x , F_y , F_z and moments M_x , M_y , M_z being exerted between the two fastening elements **11'**, **12'** may, with appropriate calibration, be calculated.

REFERENCE LIST

(21) **10** Device **10'** Device **11**, **11'** First fastening element **12**, **12'** Second fastening element **15** Length element **15a** First end **15b** Second end **20** Lever element **20a** First end **20b** Second end **21** First lever arm **21a** First lever arm end **22** Second lever arm **22a** Second lever arm end **30** Material measure **40** Scanning element **50** Second element **50a** First end **50b** Second end **51** First arm **51a** First end **51b** Second end **52** Second arm **52a** First end **52b** Second end **60** Evaluation unit **L** Length ΔL Change in length **R** Longitudinal direction **F** Force **M** Moment **D** Fulcrum **s** Path Δs Change of path **a** First length **b** Second length **E1** First plane **E2** Second plane

Claims

1. Device (**10**, **10'**) for measuring a change in length (ΔL) with a first fastening element (**11**, **11'**), a second fastening element (**12**, **12'**) and at least one length element (**15**) arranged between the two fastening elements (**11**, **11'**, **12**, **12'**) with a first end (**15a**), a second end (**15b**) and a length (**L**) along a longitudinal direction (**R**), wherein a force (**F**) acting parallel to the longitudinal direction (**R**) is configured to lead to a change in length (ΔL) of the length element (**15**), characterized in that, transversely to the longitudinal direction (**R**), a lever element (**20**) with a first end (**20a**), a second end (**20b**) and a fulcrum (**D**) is arranged, wherein the lever element (**20**) comprises a first lever arm (**21**) with a first length (**a**) between the fulcrum (**D**) and a first lever arm end (**21a**) and a second lever arm (**22**) with a second length (**b**) between the fulcrum (**D**) and a second lever arm end (**22b**), wherein the second length (**b**) is larger than the first length (**a**), wherein the length element (**15**) with its first end (**15a**) is pivotably arranged at the first lever arm end (**21a**) of the first lever arm (**21**) and wherein the second lever arm end (**22b**) of the second lever arm (**22**) is connected to a material measure (**30**), the movement of which material measure (**30**) is detectable by a scanning element (**40**).
2. Device according to claim 1, characterized in that the fulcrum (**D**) is arranged between the first end (**20a**) and the second end (**20b**) of the lever element (**20**) and the first lever arm end (**21a**) of the first lever arm (**21**) forms the first end (**20a**) of the lever element (**20**) and the second lever arm end (**22b**) of the second lever arm (**22**) forms the second end (**20b**) of the lever element (**20**).
3. Device according to claim 1, characterized in that the fulcrum (**D**) is arranged at the first end

(20a) of the lever element (20).

4. Device according to claim 1, characterized in that a second element (50) has a first end (50a) and a second end (50b) and is arranged parallel to the lever element (20) by means of a first arm (51) and a second arm (52), wherein the first arm (51) and the second arm (52) are aligned parallel to each other, wherein the second element (50) and the first arm (51) are pivotally connected to one another, wherein the second element (50) and the second arm (52) are pivotally connected to one another, wherein the lever element (20) and the first arm (51) are pivotally connected to one another, and wherein the lever element (20) and the second arm (52) are pivotally connected to one another.

5. Device according to claim 4, characterized in that the material measure (30) is arranged on the second arm (52) and extends between the lever element (20) and the second element (50).

6. Device according to claim 4, characterized in that the first arm (51) is arranged between the fulcrum (D) and the first end (50a) of the second element (50) and the second arm (52) is arranged between the second end (20b) of the lever element (20) and the second end (50b) of the second element (50).

7. Device according to claim 4, characterized in that the pivotable connections between the second element (50) and the first arm (51), the second element (50) and the second arm (52), the lever element (20) and the first arm (51), and the lever element (20) and the second arm (52) are implemented in the manner of a film hinge.

8. Device according to claim 1, characterized in that the first fastening element (11, 11') is implemented disc-shaped with a first plane (E1) and the second fastening element (12, 12') is implemented disc-shaped with a second plane (E2), wherein the first plane (E1) and the second plane (E2) are arranged parallel to each other and preferably the longitudinal direction (R) is arranged perpendicular to the planes (E1, E2).

9. Device according to claim 1, characterized in that several, in particular six, length elements (15) are arranged between the first fastening element (11') and the second fastening element (12').

10. Device according to claim 1, characterized in that the scanning element (40) is implemented as an optical, capacitive, inductive or magnetic scanning sensor.

11. Device according to claim 1, characterized in that an evaluation unit (60) is provided, which evaluates the signals detected by the at least one scanning element (40), and in particular calculates the forces and moments being exerted between the two fastening elements by means of the changes in length.
